

IDRIS RAYAN CHAKIB

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Education

University of M'Hamed Bougara Boumerdes

Boumerdes, Algeria

Master of Science in Software Engineering and Data Processing

Sep 2024 - June 2026

- **Thesis Title:** "AI Pipeline for Genetic Variant Analysis and Virtual Therapeutic Screening" (In partnership with Institut Pasteur d'Algérie).
- **Academic Distinctions:** Grade : 18/20 (Graduated with Highest Honors / Congratulations of the Jury).
- **Startup Path (Decree 1275):** Developed the project's core framework as an innovative startup. Grade: 19/20 | Awarded the Official National Startup Certificate.
- **Key Coursework:** Network and communication protocols, Cybersecurity, Distributed Systems, Software Architecture, Data Mining.

University of M'Hamed Bougara Boumerdes

Boumerdes, Algeria

Bachelor of Science in Software Engineering and Information Systems

Sep 2021 - June 2024

- **Academic Standing:** Final Grade : 10.98/20 | Class Rank: 39 / 182 (Top 21.4% of the cohort / Equivalent to ECTS Grade B).
- **Capstone Project Title:** "Fraud Monitoring Web Solution with Machine Learning" (In partnership with Ooredoo Algeria).
- **Project Distinction:** Grade 16/20.
- **Key Coursework:** Object-Oriented Programming, Calculus, Linear Algebra, Probability & Statistics, Databases.

Research Publication

- Benhenni O., Gaceb D., Touazi F., Rouibi B., Ismail A., Kebour W., **Idris R. C.**, Galleze A., Benidir M. "SmartADN: An End-to-End Deep Learning Pipeline for Variant Pathogenicity Classification and Mutation-Aware Therapeutic Screening in the North African Context." Submitted to the IEEE International Symposium on iNnovative Informatics of Biskra (IEEE ISNIB'26), 2026. (Paper ID: 101 Currently Under Peer-Review / Forthcoming).
- Benhenni O., Gaceb D., Touazi F., Kebour W., **Idris R. C.**, Rouibi B., Ismail A. "An End-to-End Approach for SNV Pathogenicity Prediction via Progressive Fine-Tuning of Nucleotide Transformer-v2." Published and presented at the 3rd National Conference on Applications of Artificial Intelligence (A2I'26), 2026. (Official Certificate of Participation Awarded as **Presenting Author**).

Professional Experience

Institut Pasteur d'Algérie

Algiers, Algeria

Data Processing & AI Research Intern

Feb 2026 – June 2026

- Co-designed SmartADN, an integrated pipeline coupling a sequence-only pathogenicity classifier with a mutation-aware therapeutic screening module.
- Fine-tuned the **Nucleotide Transformer-v2** genomic foundation model (500M parameters) using a layer-wise unfreezing strategy under a strict gene-split out-of-distribution (OOD) protocol.
- Achieved a near-perfect AUC-ROC of **0.987** and **94%** overall accuracy on entirely held-out clinically critical genes (BRCA1, BRCA2, TP53) without relying on population allele-frequency annotations.
- Engineered the protein backbone integration for **AEGIS-GT**, mapping conformational perturbations via the pointwise difference of ESM-2 embeddings ($\Delta e = eMUT - eWT$).
- Implemented a **bi-phasic training strategy** using a linear zero-initialised adapter to successfully bridge biophysical Silver-Standard descriptors with experimental Gold-Standard affinity data.

Ooredoo Algeria

Algiers, Algeria

Software Engineering Intern

Feb 2024 – June 2024

- Developed predictive machine learning models integrated into the enterprise telecom web infrastructure to track fraudulent patterns.
- Analyzed large telecom datasets to improve data extraction pipelines, increasing tracking and software reliability.

Projects

Multi-Robot Autonomous 2D Swarm Exploration

- Engineered a multi-robot swarm mapping system from scratch using a continuous Teacher-Student reinforcement learning strategy paired with the A* pathfinding algorithm.
- Coded the entire Deep Q-Network (DQN) architecture, backpropagation loops, and neural layers using raw Python/NumPy.

Nexus: Decentralized P2P Encrypted Chat System

- Designed a robust peer-to-peer distributed framework featuring dynamic private chat rooms, logical isolation, and multi-threaded file transfer protocols.
- Implemented physical concurrency controls and data consensus layers using Lamport Logical Clocks, custom Byzantine-resilient voting systems, and multi-layered Fernet cryptographic encryption via Tkinter GUI.

Mooglee: Natural Language Processing Search Engine

- Built a highly optimized localized text search engine integrating foundational Natural Language Processing (NLP) preprocessing pipelines.
- Developed custom algorithmic structures for tokenization, automated stop-word elimination, Porter-style stemming/unstemming, and contextual Bi-gram language modeling.

Leadership Experience

SmartADN Startup Project

Boumerdes, Algeria

Co-Founder & Lead AI Engineer

Sept 2025 – Present

- Selected for the prestigious Algerian National Student-Entrepreneur program under Ministerial Decree 1275, demonstrating academic and technical leadership.
- Architected and developed AEGIS-GT, a custom deep learning model designed for advanced genomic analysis and core startup technology.
- Successfully pitched the technical design and AI model architecture to academic committees, securing the official National Startup Certificate.

NASA International Space Apps Challenge

Global / Algiers Cohort

Team Leader & Core Developer

- Led a multidisciplinary team of student engineers and peers during the 48-hour global hackathon to address complex astronomical challenges.
- Co-designed and engineered a 3D interactive web-based Orrery application to simulate celestial body trajectories using open-source NASA telemetry data.
- Managed project workflows, delegated tasks among university peers under strict time constraints, and delivered the functional backend framework, awarded the **Galactic Problem Solver Certificate**

Additional Credentials & Activities

Professional Certifications

2025 – 2026

- **Machine Learning Specialization (DeepLearning.AI):** Certified core expertise in neural networks and AI system design.
- **Google Data Analytics Professional Certificate:** Certified proficiency in data cleaning, SQL analysis, and problem-solving.

Active Competitor, Codeforces Platform

2026 – Present

- Current Rating: 1166 (Actively progressing toward top-tier algorithmic milestones via timed challenges.)
- Contestant, ICPC Online Challenge 1 powered by Huawei, in which i developed a custom scheduling algorithm to optimize concurrent request routing for Edge-Cloud Collaborative LLM Inference loops.

Technical Skills

- **Languages:** Java, Python, C/C++, SQL, JavaScript, HTML/CSS.
- **Frameworks:** Next, Express, FastAPI, Flask, Pytorch, Keras, Scikit-learn, Tensorflow, Hugging Face (Transformers).
- **Developer Tools:** Git, Docker, Node.js, VS Code, Visual Studio, Pycharm, IntelliJ, Eclipse.
- **Libraries:** React, pandas, Numpy, Matplotlib, seaborn.